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SIEM SECURITY ANALYSIS REPORT

Generated: 2026-03-31 23:26:49 | Log Source Type: Postfix

Total Events	1000
Parsed Successfully	1000 (100%)
Risk Level	MEDIUM
ML Detected Threats	0

Overall Risk Score: 40/100 (MEDIUM)

1. EXECUTIVE SUMMARY

This comprehensive security report presents the analysis of **1,000** log events from the SIEM system. The analysis successfully parsed **1,000** events (100% success rate) with **0** events failing to parse.

■ **No ML-detected threats** were found in the analyzed log sample. The security posture appears normal.

2. AI-POWERED THREAT ANALYSIS

This section contains detailed threat analysis generated by our AI SOC Analyst.

SIEM Security Analysis Report

Date: March 19, 1900 (Based on log data)

Analyst: AI Security Analyst

1. EXECUTIVE SUMMARY:

The organization's email infrastructure is currently under potential attack, exhibiting signs of brute-force attempts, threat intelligence matches indicating malicious activity, and suspicious spam activity. Immediate investigation and mitigation steps are required to prevent further compromise and potential data breaches.

2. KEY FINDINGS:

| Finding | Severity | Description |

|---|---|---|

| Multiple Brute-Force Attacks | HIGH | Detected brute-force attempts against the email server, potentially targeting user credentials. |

| Threat Intelligence Match | HIGH | Identified malicious activity matching known threat intelligence indicators, suggesting a targeted attack. |

| Suspicious Spam Activity | MEDIUM | High volume of outgoing emails or emails with suspicious characteristics detected, potentially indicating a compromised account or spam campaign. |

| High Connection Rate from a Single IP | MEDIUM | IP address 45.115.119.123 exhibited a high connection rate (11/3600s), potentially indicative of automated scanning or brute-force activity. |

| Internal Relay Error | LOW | An error log indicates a potential issue with internal mail relay configuration, which could be exploited if not properly secured. |

3. ATTACK CHAIN ANALYSIS:

Based on the alerts and log data, a potential attack chain can be inferred:

* **Reconnaissance (TA0043):** The high connection rate from IP 45.115.119.123 suggests potential scanning or reconnaissance activity targeting the email server.

- * **Credential Access (TA0006):** The detected brute-force attacks indicate attempts to gain unauthorized access to user accounts by guessing passwords.
- * **Initial Access (TA0001):** Successful brute-force attacks or exploitation of vulnerabilities could provide initial access to the email system.
- * **Execution (TA0002):** If initial access is gained, attackers could execute malicious code, potentially through phishing emails or exploiting vulnerabilities in email attachments.
- * **Exfiltration (TA0010):** Once inside, attackers may attempt to exfiltrate sensitive data from the email system.
- * **Command and Control (TA0011):** A compromised system could be used as a command and control server for further malicious activities.

4. THREAT INTELLIGENCE:

- * **IOCs (Indicators of Compromise):**
 - * IP Address: 45.115.119.123 (High connection rate, potential brute-force source)
 - * Email Address: sujithonrao931@gmail.com (Appears in relayed inbound mail)
- * **Suspicious IPs:**
 - * 209.85.210.52, 209.85.161.69 (High log count, requires further investigation)
- * **Attacker TTPs (Tactics, Techniques, and Procedures):**
 - * Brute-forcing email accounts (T1110)
 - * Potential spamming activity (T1486)
 - * Scanning for vulnerabilities (T1595)

5. RISK ASSESSMENT:

Current Risk Level: HIGH

Justification: The combination of brute-force attacks, threat intelligence matches, and suspicious spam activity indicates a significant risk to the organization's email infrastructure. A successful attack could lead to compromised accounts, data breaches, financial losses, and reputational damage. The high connection rate from a single IP further increases the risk, suggesting a targeted and persistent attack.

6. RECOMMENDATIONS:

- * **Implement Multi-Factor Authentication (MFA) (Priority: HIGH):** Enforce MFA for all email accounts to significantly reduce the risk of successful brute-force attacks.

2. ****Block Suspicious IP Addresses (Priority: HIGH):**** Immediately block IP address 45.115.119.123 and any other IPs identified as malicious in the firewall and email security gateway.

3. **Investigate Threat Intelligence Matches (Priority: HIGH):**** Conduct a thorough investigation of the threat intelligence matches to determine the scope and impact of the potential attack.**

4. ****Review and Harden Email Security Configuration (Priority: MEDIUM):**** Review and harden the email security configuration, including spam filtering, anti-phishing measures, and attachment scanning.

5. ****Implement Rate Limiting (Priority: MEDIUM):**** Implement rate limiting on email connections to prevent brute-force attacks and other malicious activities.

6. ****Monitor for Suspicious Outgoing Email (Priority: MEDIUM):**** Implement monitoring rules to detect and alert on suspicious outgoing email activity, such as high volumes of emails or emails with unusual characteristics.

7. ****Patch Management (Priority: MEDIUM):**** Ensure all email servers and related systems are up-to-date with the latest security patches to address known vulnerabilities.

****7. INCIDENT RESPONSE:****

1. ****Containment:****

- * Isolate any potentially compromised accounts or systems.
- * Block malicious IP addresses and domains.
- * Disable compromised user accounts.

2. ****Eradication:****

- * Remove any malware or malicious code from compromised systems.
- * Reset passwords for all affected user accounts.
- * Review and update security policies and procedures.

3. ****Recovery:****

- * Restore systems from backups if necessary.
- * Monitor systems for any signs of further compromise.
- * Communicate with stakeholders about the incident and the steps taken to resolve it.

4. ****Post-Incident Activity:****

- * Conduct a thorough post-incident review to identify the root cause of the incident and any lessons learned.
- * Update security policies and procedures based on the findings of the review.
- * Implement additional security measures to prevent future incidents.

This report provides a preliminary analysis of the security events. Further investigation and analysis may be required to fully understand the scope and impact of the potential attacks.

3. THREAT INTELLIGENCE & IOCs

Top Source IP Addresses (by event count)

Rank	IP Address	Events	Risk Assessment
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

Most Targeted User Accounts

Rank	Username	Events	Priority
1	iqac@tmv.edu.in	18	MEDIUM
2	infocenter@tmv.edu.in	14	MEDIUM
3	finance@tmv.edu.in	11	MEDIUM
4	0.001	10	MEDIUM
5	examination@tmv.edu.in	10	MEDIUM
6	kulguru@tmv.edu.in	8	MEDIUM
7	law@tmv.edu.in	8	MEDIUM
8	administration@tmv.edu.in	8	MEDIUM
9	phd@tmv.edu.in	6	MEDIUM
10	tmvcompscience@gmail.com	6	MEDIUM

5. SECURITY ALERTS

Total alerts generated: 4. Showing top 15 alerts.

Alert #1: bruteforce

Timestamp: 1900-03-18T16:11:13 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

Alert #2: threat_intel

Timestamp: 1900-03-18T16:28:39 | **Source:** N/A

Message: Attack detected: threat_intel (confidence: 80%)

Alert #3: spam_activity

Timestamp: 1900-03-18T16:29:07 | **Source:** N/A

Message: Attack detected: spam_activity (confidence: 60%)

Alert #4: bruteforce

Timestamp: 1900-03-18T16:34:43 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

6. LOG ANALYTICS & STATISTICS

6.1 Severity Distribution

Severity Level	Event Count	Percentage
CRITICAL	0	0.0%
ERROR	6	0.6%
WARNING	7	0.7%
INFO	987	98.7%
DEBUG	0	0.0%

6.2 Log Type Distribution

Log Type	Count
postfix	1000

6.3 Top Source IP Addresses

Rank	IP Address	Event Count	Activity Level
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

7. RECOMMENDATIONS & REMEDIATION

◆ Critical

Immediately investigate all critical and high-severity alerts. Isolate affected systems if compromise is suspected.

◆ High Priority

Review and block suspicious IP addresses with high event counts. Implement additional monitoring on targeted accounts.

◆ Medium Priority

Enable enhanced logging for critical systems. Review access controls for accounts with failed authentication attempts.

◆ Best Practices

Ensure all systems are patched and up-to-date. Implement MFA for all privileged accounts. Regular review of security logs.

◆ Monitoring

Set up real-time alerts for threshold-based detections. Create correlation rules for multi-stage attack patterns.

◆ Response

Document and test incident response procedures. Ensure backup and recovery mechanisms are functional.

8. APPENDIX

Report Metadata

Report Generated	2026-03-31 23:26:49
Log Source Type	Postfix
Total Lines	0
Parsed Lines	0
Failed Lines	0
ML Attacks Detected	0
Total Alerts	4
Analysis Engine	Cyber Chakshu SIEM v2.0 + AI SOC Analyst

Disclaimer: This report is generated automatically by Cyber Chakshu SIEM system. The analysis is based on log data and machine learning models. Human review is recommended for critical security decisions.